Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

The concepts from the class that I expect to use are conditionals, lists, loops, dataclasses, random, dictionary, dataclasses, nested data and the assert_equals test. I want to improve my debugging skills while completing this project.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

The easiest part is going to be naming functions and their conditionals because I understand tennis very well. It is mainly if statements. The hardest part in my opinion is going to be getting drafted to work and making sure everything is in proper order when it comes to routes and data classes.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

The parts that are going to require the most time and effort are calculating serve percentage and creating a history list, debugging written tests, and setting up the Drafter pages in routes with State passing between them.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Distractions would be my phone, work, studying for work, gym, socializing. I will plan to block time each day to commit to doing work for this project. The things I will need are Drafter and Bakery documentation along with my old Drafter project and class notes. I do not really have any questions about the assignment.

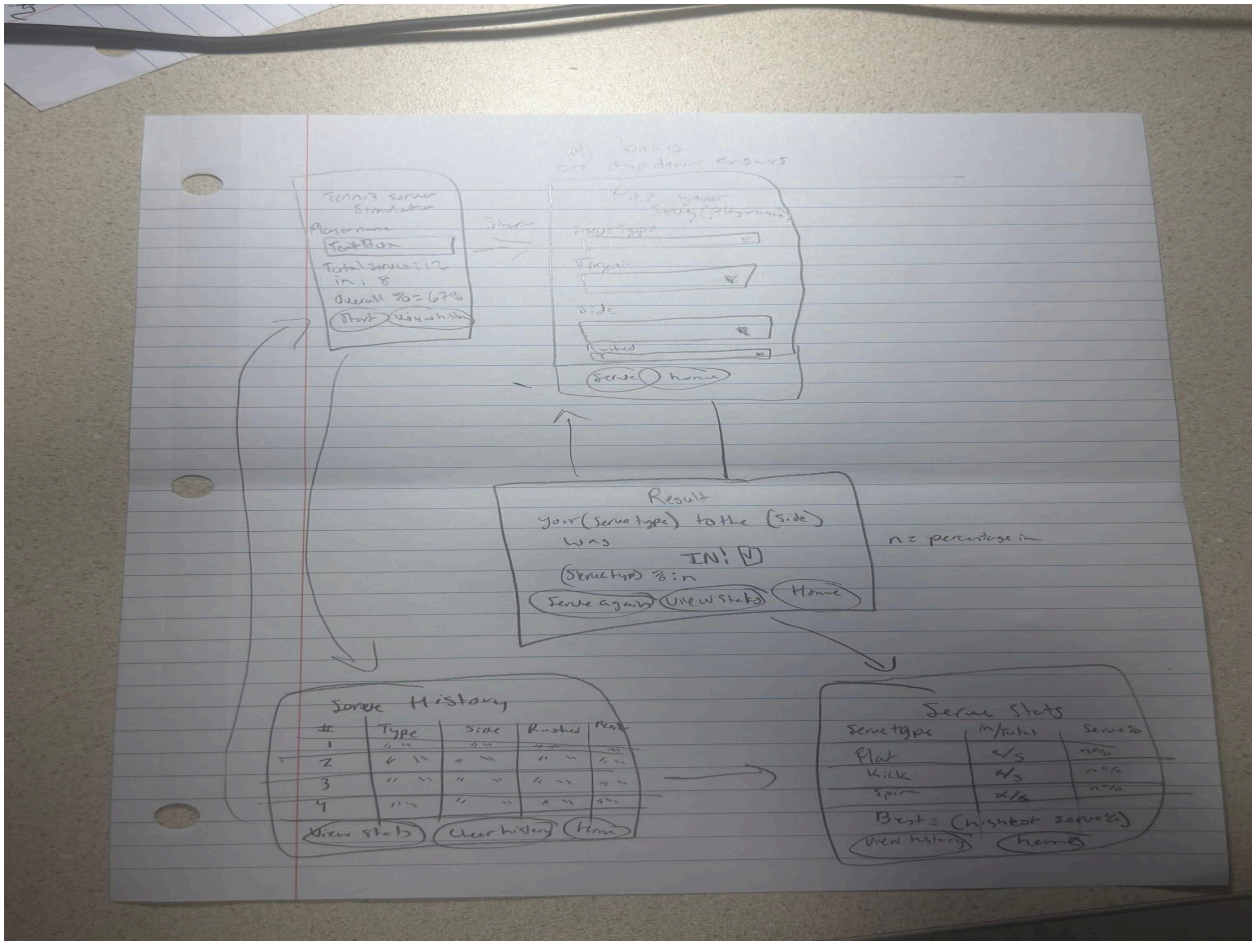
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


Result
your (serve type) to the (side)
was
IN!
n = percentage in
Serve again
View Stats
Home
Serve History
#
Type
Size
Result
Clear history
Serve Stats
Serve type
% / total
Flat
Kick
Spin
Home

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	A user picks a serve type, target, and side, and the site tells them if it went in or out. It tracks their serve percentage allowing for the most reliable to be displayed.
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	● player_name: str: The name typed on the home page, It is shown on the server page ● history: list[Serve]: Logs every hit serve. A new Serve is added to hit_serve, and the list empties through clear_history ● total_serves: int: Goes up 1 after every serve. Back to 0 when history is cleared ● serves_in: int: goes up when the ball lands in. Moves serve % up.
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	● serve_type: str (“Flat”, “Kick”, or “Spin”) ● Target: str (“T”, “Body”, “Wide”) ● Side: str (“duece”, “ad”) ● Rushed: bool (rushed the service motion) ● Landed_in: bool (landed inside the service box on the right side)
Routes (minimum 7) The names of your route functions	Index, set_name, serve_form, hit_serve, view_history, view_stats, clear_history

Pages (minimum 5) A brief description of the pages in your site	Home: place where the name box, total serves, and overall serve exist % Serve form: page where you pick the serve type, target, side, and if it was rushed Result: determines if it is in or out and increases serve % for that serve type History: a log with every serve attempted Stats: serve % for each type and the best serve based on highest serve %
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	TextBox for player name SelectBox for serve type SelectBox for target SelectBox for side CheckBox for “rushed”
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	If rushed is True, -15% from probability of serve going in If total_serves is 0, show “No serves yet! Instead of dividing by 0 to find average If the served type matches the type being counted, add it to the type's total, and add it to that type’s “in” count if it landed inside of the box.
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	A for loop will go over state.history and count the total amount of serves and the serves that went in. That will give a serve % for each type of serve displayed on the Stats and Result pages. Based on this information, it will pick the best serve and build rows in a history table.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I did not use an LLM

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.